XINRAN ZHANG

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PROFILE

My research focuses on human–environment relationships through GIScience, spatial analysis, and computational methods. I am particularly interested in developing interpretable and reusable computational methods for understanding spatial and temporal relationships, bridging methodological innovation with practical research applications.

EDUCATION

University of Edinburgh, Edinburgh College of Art **Sep 2023-Sep 2024**

MSc in Landscape and Wellbeing (Merit) *Edinburgh, UK*

Dissertation: Soundscape Perception and Social Interaction in Urban Blue Spaces: A Mixed-Methods Study (82/100, Distinction)

Fuzhou University, School of Architecture and Urban Planning **Sep 2017-Jun 2022**

BEng in Landscape Architecture *Fuzhou, CN*

Undergraduate Thesis: GIS-Based Noise Mapping and Soundscape Informed Urban Park Design (Top 2 in cohort)

RESEARCH EXPERIENCE

Street-View-Based Environmental Representation: StreetSignals & Beyond Semantics **2025**

Independent Research | StreetSignals | Beyond Semantics

- Investigated how conventional semantic indicators may overlook spatial configurations relevant to human environmental perception and exposure assessment.
- Developed a semantic-segmentation workflow using 200 street-view images to derive interpretable indicators of traffic dominance and pedestrian-relevant environmental conditions.
- Proposed a structure-aware representation framework extending beyond semantic composition to consider spatial continuity, connectivity, and coherence as complementary dimensions of environmental assessment.

HumanRoute: Profile-Conditioned Pedestrian Environment Assessment with Vision–Language Models **May 2026-Present**

Independent Research | Project Overview

- Investigated how environmental assessments may vary with users' mobility capabilities and the spatial context available to vision-language models.
- Developed a VLM evaluation pipeline using Mapillary imagery from 13 Seattle pedestrian routes (~8 km), with multi-scale sampling (12/24/48 m) and contextual inputs (3/5/9 frames).
- Designed six-dimensional, profile-conditioned assessments for adults, older adults, and wheelchair/stroller users to examine differences in accessibility-related judgments.

Blue-Space Soundscape Restoration and Social Behaviour: A Mixed-Methods Study **Mar 2024-Aug 2024**

Master's Dissertation, Supervised by Prof. Simon Bell | Distinction (82/100) | Project Overview *Glasgow, UK*

- Investigated how physical and perceptual characteristics of urban blue spaces relate to psychological restoration and social behaviour, with the aim of identifying potential mechanisms relevant to evidence-based spatial improvements.
- Conducted environmental and soundscape assessments across 65 locations, recorded 1,780 behavioural observations, and interviewed 55 participants.
- Integrated spatial exposure mapping, regression analysis, and qualitative interpretation to examine relationships between environmental conditions, staying behaviour, and social interaction, including age-related differences in perception.

AFTERMATH: Evaluating Terminal-State Externalities in Embodied Spatial Planning **Jul 2026–Present**

Independent Researcher & Developer | GitHub | Project Overview

- Developed a reproducible spatial-planning benchmark to investigate how robot-efficient decisions can create unequal accessibility outcomes for subsequent users with different mobility requirements.
- Evaluated ~1,220 planning episodes across 20 scenes and three mobility profiles; identified accessibility harm in 18/20 robot-efficient baseline states and improvements in eight scenes within a 15% robot-cost allowance.

RESEARCH SOFTWARE & TOOL DEVELOPMENT

Grain Checker: QGIS Plugin for MAUP Sensitivity Diagnostics **Feb 2026-Mar 2026**

Research Tool Development (Independent) |Github| Demo video

- Developed a QGIS plugin to assess the sensitivity of bivariate spatial relationships to aggregation scale, supporting reproducible diagnostics of the modifiable areal unit problem (MAUP).

Parcel-Level Land-Use Continuity: A QGIS Tool for Life-Course Exposure Reconstruction **Apr 2025-Jul 2025**

Research Tool Development (Independent) |Github | Demo video:

- Developed a QGIS tool to reconstruct historical parcel-level land-use continuity across changing spatial boundaries, supporting longitudinal environmental exposure analysis.

TECHNICAL & ANALYTICAL SKILLS

GIScience & Spatiotemporal Analysis: Spatial statistics, spatial interpolation (kriging), spatial accessibility analysis, longitudinal land-use reconstruction, historical parcel matching, spatial aggregation and MAUP sensitivity analysis, space syntax, geospatial data processing.

Geospatial Computing & Research Software: Python (NumPy, geospatial data processing, algorithm implementation), QGIS plugin development, GIS workflow automation, reproducible computational experiments, graph-based pathfinding (A*), computational geometry.

GeoAI & Spatial Data Representation: Street-view semantic segmentation (ResNet50), computer vision, vision-language model evaluation, multimodal prompting, spatial structural indicators, multi-scale sampling, geospatial visualization.

Empirical Research & Software Tools: Behavioral observation, semi-structured interviews, mixed-methods analysis; QGIS, ArcGIS, R, Stata, NVivo, AnyLogic, Cadna/A, PyTorch, Git/GitHub.

AWARDS & CONFERENCE PRESENTATIONS

National Award & Oral Presentation, 2nd National Soundscape Academic Forum of China **Oct 2022**

Best Undergraduate Project Award (Top 6%), School of Architecture and Urban Planning of FZU **Jun 2022**

National Prize (Top 2.6%, 18/784 entries), National Urban-Rural Planning Competition for Undergraduates **Dec 2020**

PROFESSIONAL EXPERIENCE

Research Assistant, Fuzhou University & Collaborative Labs **Sep 2019-Mar 2024**

SOCIAL ENGAGEMENT

Independent knitwear designer, Ravelry **2026**

Student Advisor, Design Competitions, School of Architecture and Urban Planning, FZU **Fuzhou, CN, 2021**

Volunteer, Welfare Home for Individuals with Cognitive Impairments **Fuzhou, CN, 2017-2019**

- Supported daily services and developed early awareness of sensory processing differences and accessibility needs.